STAT 6900: Stochastic Processes in Modern Machine Learning
Lecture 5: Reinforcement Learning in Continuous Time and Space

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Discrete-Time RL Essentials

- **MDP:** $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$, with $S_{t+1} \sim P(\cdot \mid S_t, A_t)$.
- **Policy:** $\pi(a \mid s)$ (deterministic: $a = \pi(s)$; stochastic: $a \sim \pi(\cdot \mid s)$).
- **Return:** $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$.
- **Value functions:**

$$v^\pi(s) = \mathbb{E}^\pi[G_t \mid S_t = s], \qquad q^\pi(s,a) = \mathbb{E}^\pi[G_t \mid S_t = s, A_t = a].$$

- **Bellman expectation equation:**

$$v^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a)\big(r(s,a,s') + \gamma v^\pi(s')\big).$$

- **Optimality:**

$$v^*(s) = \max_a \sum_{s'} P(s' \mid s, a)\big(r(s,a,s') + \gamma v^*(s')\big),$$

$$\pi^*(s) \in \arg\max_a (\cdots).$$

RL in Continuous Time: What Has Been Studied

- Many real systems evolve in **continuous time** (e.g., trading, robotics, driving), motivating continuous-time RL.
- Classical optimal control/stochastic control already studies **controlled diffusions** via SDEs and HJB equations.
- In RL, **exploration** is typically modeled by **randomizing actions** (stochastic policies) in discrete time.
- Extending stochastic policies to continuous time is **non-trivial**: action randomization must occur **continuously**.
- Prior continuous-time approaches exist, but a rigorous framework that **incorporates exploration into the state dynamics** and links it to **stochastic control theory** is less developed.

Goal of Lecture 6

- **Main goal:** provide a **stochastic control** framework for RL in **continuous time and space**.
- **Core innovation:** propose an **exploratory formulation** of the state dynamics that captures **learning under exploration** in the continuous-time limit.
- **Key modeling move:** exploration enlarges controls from **Dirac measures** (pure actions) to **probability distributions** over actions (relaxed controls).
- **Illustration objective:** study the **exploration–exploitation trade-off** using an **entropy-regularized** reward.

Mathematical Tools We Will Use

- **Controlled SDEs:** state evolution under control (drift/volatility depend on control).
- **Itô calculus:** Itô's formula, quadratic variation, and stochastic integration.
- **Infinitesimal generator:** connects the SDE to PDE/HJB (local Bellman principle).
- **Dynamic Programming Principle (DPP):** continuous-time Bellman idea $\Rightarrow$ HJB.
- **Hamilton–Jacobi–Bellman (HJB) equation:** characterize the value function and the optimal control.
- **Relaxed controls / measure-valued controls:** controls are **distributions** over actions.
- **Entropy regularization:** Shannon differential entropy encourages exploration and yields a tractable optimizer.
- **Linear–Quadratic (LQ) structure:** solvable case giving explicit optimal distributions (Gaussian).

Discrete vs Continuous RL

Discrete-Time RL	**Continuous-Time RL**
State: $S_t \in \mathcal{S}$	State: $X_t \in \mathbb{R}^n$
Action: $A_t \in \mathcal{A}$	Action/control: $u_t \in U \subseteq \mathbb{R}^m$
Dynamics: $S_{t+1} \sim P(\cdot \mid S_t, A_t)$	Dynamics: $dX_t = b(X_t, u_t)\,dt + \sigma(X_t, u_t)\,dW_t$
Policy: $\pi(a \mid s)$	Control law: $u_t = u(t, X_t)$ (feedback)
Exploration: stochastic $\pi(a \mid s)$	Exploration: distribution-valued control $\pi_t(\cdot) \in \mathcal{P}(U)$
Bellman equation	HJB equation (continuous-time Bellman)
Value: v^π, q^π	Value: V^π (often PDE characterized)
Optimization: $\max_a(\cdots)$	Optimization: $\sup_{u \in U}(\cdots)$ or $\sup_{\pi \in \mathcal{P}(U)}(\cdots)$

- **Key message:** Bellman $\Rightarrow$ HJB; action randomization $\Rightarrow$ relaxed (distributional) control.

Exploratory Formulation: Probabilistic Setup

- We begin with a filtered probability space
$$(\Omega, \mathcal{F}, \mathbb{P}; \{\mathcal{F}_t\}_{t\geq 0}),$$
together with an $\{\mathcal{F}_t\}_{t\geq 0}$-adapted Brownian motion
$$W = \{W_t,\ t \geq 0\}.$$
- The **action space** U represents the set of all feasible decisions available to the agent.
- An admissible **open-loop control** is a process
$$u = \{u_t,\ t \geq 0\}$$
such that
  - u_t is $\mathcal{F}_t$-adapted,
  - u_t is measurable,
  - $u_t \in U$ for all $t \geq 0$.
- Interpretation:
  - u_t is the action chosen at time t,
  - the choice can depend on currently available information,
  - but it is not yet written explicitly as a function of the state.

Classical Controlled State Dynamics

- In the classical stochastic control setting, the state (or feature) process evolves according to

$$dx_t^u = b(x_t^u, u_t)\,dt + \sigma(x_t^u, u_t)\,dW_t, \qquad t > 0,$$

with initial condition

$$x_0^u = x \in \mathbb{R}.$$

- Here:
  - x_t^u is the state under the control u,
  - $b(x, u)$ is the drift term,
  - $\sigma(x, u)$ is the diffusion coefficient,
  - x denotes the current state of the system.
- This is the continuous-time analogue of a controlled Markov system in discrete time:

$$X_{t+1} \sim P(\cdot \mid X_t, A_t).$$

- Main point:
  - the control affects the state dynamics,
  - the state remains random because of the Brownian noise.

Classical Objective and Value Function

- The goal is to maximize the expected total discounted reward:

$$V^{cl}(x) := \sup_{u \in \mathcal{A}_{cl}(x)} \mathbb{E}\left[\int_0^\infty e^{-\rho t} r(x_t^u, u_t)\, dt \;\middle|\; x_0^u = x \right].$$

- Here:
  - $r(x, u)$ is the instantaneous reward,
  - $\rho > 0$ is the discount rate,
  - $\mathcal{A}_{cl}(x)$ is the set of admissible controls.
- Interpretation:
  - reward is accumulated over the whole time horizon,
  - future rewards are discounted by $e^{-\rho t}$,
  - the optimization is over all admissible control processes.
- This is the continuous-time counterpart of the discrete-time value function:

$$v^\pi(s) = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \;\middle|\; S_t = s \right].$$

Classical Feedback Control When the Model Is Known

- In the classical setting, the model is assumed to be known:

$$b, \qquad \sigma, \qquad r$$

  are specified.
- Then dynamic programming can be applied, and the optimal control can be represented as a deterministic feedback rule

$$u_t^* = \mathbf{u}^*(x_t^*).$$

- Here:
  - $\mathbf{u}^*$ is a deterministic mapping from the current state to the action space U,
  - x_t^* is the state process under the optimal control.
- Interpretation:
  - once the optimal rule $\mathbf{u}^*$ is derived at the initial stage,
  - it is implemented for all $t \geq 0$ by observing the current state and applying

$$u_t^* = \mathbf{u}^*(x_t^*).$$

Why RL Requires Exploration

- In reinforcement learning, the underlying model is **not known**.
- Therefore, the agent cannot directly compute the optimal feedback law from

$$b, \ \sigma, \ r,$$

  because these ingredients are not fully available.
- Instead, the agent must **learn** the environment dynamically.
- This requires **exploration**:
  - the agent interacts with the environment,
  - tries different actions,
  - and improves decisions through trial and error.
- Hence the main modeling challenge is:
  - how do we represent exploration mathematically in continuous time?

Key Idea: Exploration Through Control Distributions

- We model exploration by introducing a **distribution of controls**

$$\pi = \{\pi_t(u),\ t \geq 0\}$$

  over the action space U.
- At each time t, the control is not taken as one fixed value only.
- Instead, the agent chooses a distribution π_t, and each trial samples a classical control from this distribution.
- Thus, the notion of control is extended from: $u_t \in U$ to $\pi_t(\cdot) \in \mathcal{P}(U)$.
- Interpretation:
  - deterministic or classical control: choose one action,
  - exploratory control: choose a probability law over actions.
- This is the first major step from classical stochastic control toward RL in continuous time.

Repeated Trials and the N-Round Interpretation

- To motivate the exploratory formulation, we consider the following experiment:
  - the agent executes the control over N rounds,
  - all rounds use the same time horizon,
  - in each round, a classical control is sampled from the distribution π.
- So the agent repeatedly interacts with the same control problem under the same exploratory policy.
- When N is large, the reward generated by this policy becomes statistically accurate.
- This repeated-learning viewpoint motivates replacing a single sampled action by an **averaged effect** under the control distribution.
- This is exactly what leads, in the next step, to the **exploratory state dynamics** and the **exploratory reward**.

Policy Evaluation in Continuous Time

- We first consider the special case where the reward depends only on the control:
$$r(x_t^u, u_t) = r(u_t).$$
- To evaluate a control distribution π, we consider N identical independent copies of the control problem.
- At round i, a classical control
$$u^i = \{u_t^i,\ t \geq 0\}$$
is sampled under the (possibly random) control distribution π.
- Each sampled control is then applied to its own copy of the original classical control problem.
- This repeated-round construction is the continuous-time analogue of **policy evaluation**:
  - fix the policy $\rightarrow$ run many trials $\rightarrow$ average the observed rewards
- The goal is to identify the **effective reward** generated by the exploratory policy π.

Law of Large Numbers and Averaged Reward

- For a small interval $[t, t+\Delta t]$, the average reward over N rounds is

$$\frac{1}{N}\sum_{i=1}^{N} e^{-\rho t} r(u_t^i)\Delta t.$$

- As $N \to \infty$, the law of large numbers suggests

$$\frac{1}{N}\sum_{i=1}^{N} e^{-\rho t} r(u_t^i)\Delta t \xrightarrow{\text{a.s.}} \mathbb{E}\left[e^{-\rho t}\int_U r(u)\pi_t(u)\,du\,\Delta t\right].$$

- Interpretation:
  - the sampled controls $u_t^1, \ldots, u_t^N$ fluctuate from round to round,
  - but their average effect converges to the reward averaged under the density π_t.

Law of Large Numbers and Averaged Reward

- Hence the policy distribution π_t induces an **averaged instantaneous reward**

$$\int_U r(u)\pi_t(u)\,du.$$

- This is the first indication that randomized control in continuous time should be modeled through **integrals against the control distribution**.

Why the General Reward Case Is Harder

- The previous argument works easily when the reward depends only on the control.
- In the general RL setting, however, the reward has the form

$$r(x_t^u, u_t),$$

- so it depends on both: the chosen action/control and the current state of the system.
- Therefore, randomizing the control does **not** only affect the reward directly.
- It also changes the **state dynamics**, because the state itself evolves under the sampled control.
- This leads to the key question:
  - if controls are sampled repeatedly from a distribution π, what should the resulting continuous-time state dynamics look like?
- We answer this by defining the **exploratory state dynamics**.

Repeated Learning Under a Control Distribution

- To understand how exploration changes the state dynamics, we again consider N identical independent copies of the control problem.
- At round i:
  - a control u^i is sampled from the control distribution π,
  - the corresponding state process is denoted by $x^i = \{x_t^i,\ t \geq 0\}$,
  - and x^i solves the classical SDE under the sampled control u^i.
- Thus:
  - x_t^i is the state in the i-th independent trial,
  - u_t^i is the sampled control used in that trial.
- The idea is to study the empirical average of the state increments across rounds and then pass to the limit as $N \to \infty$.
- This will reveal the **effective drift** and **effective volatility** produced by exploration.

Averaging State Increments Over Many Rounds

- For each round $i = 1, \ldots, N$, let x_t^i be the state process generated by the sampled control u_t^i.
- Over a small interval $[t, t + \Delta t]$, consider the empirical averages

$$\frac{1}{N} \sum_{i=1}^{N} \Delta x_t^i, \qquad \frac{1}{N} \sum_{i=1}^{N} (\Delta x_t^i)^2.$$

- As $N \to \infty$, these describe the average local drift and average local variance induced by the control distribution π_t.
- This suggests that under exploration, the dynamics should depend on the distribution π_t, not on one single sampled action.

From Many Sampled Runs to One Exploratory Process

- For the i-th sampled round, the state process satisfies

$$dx_t^i = b(x_t^i, u_t^i)\,dt + \sigma(x_t^i, u_t^i)\,dW_t^i, \qquad x_0^i = x,$$

  where u_t^i is the sampled classical control in round i.
- Instead of tracking all sampled trajectories

$$x^1, \ldots, x^N,$$

  we introduce one effective state process

$$X^\pi = \{X_t^\pi,\ t \geq 0\},$$

  which represents the averaged evolution under the control distribution π.
- This effective process is defined by

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\,dt + \tilde{\sigma}(X_t^\pi, \pi_t)\,dW_t, \qquad X_0^\pi = x.$$

- Thus:
  - x_t^i: state in the i-th sampled round,
  - X_t^π: effective averaged state under π.

Exploratory Drift and Volatility

- The goal is to match the first two local moments induced by repeated learning.
- For the exploratory process,

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\, dt + \tilde{\sigma}(X_t^\pi, \pi_t)\, dW_t,$$

  we have heuristically:

$$\mathbb{E}[\Delta X_t^\pi] \approx \tilde{b}(X_t^\pi, \pi_t)\Delta t,$$

  and

$$\mathbb{E}[(\Delta X_t^\pi)^2] \approx \tilde{\sigma}^2(X_t^\pi, \pi_t)\Delta t.$$

- From the N-round argument, the corresponding empirical limits are

$$\int_U b(X_t^\pi, u)\pi_t(u)\, du \quad \text{and} \quad \int_U \sigma^2(X_t^\pi, u)\pi_t(u)\, du.$$

- Therefore the definitions of $\tilde{b}$ and $\tilde{\sigma}$ are chosen precisely to make these quantities agree.

Exploratory Drift

Definition 6.1

Exploratory drift For $y \in \mathbb{R}$ and $\pi \in P(U)$, the exploratory drift is defined by

$$\tilde{b}(y, \pi) := \int_U b(y, u)\pi(u)\, du.$$

- Interpretation:
  - $b(y, u)$ is the classical drift under action u,
  - $\pi(u)$ is the probability density assigned to u,
  - $\tilde{b}(y, \pi)$ is the **average drift** under the randomized action.
- So exploration changes the drift by replacing one action u with an average over all actions weighted by π.
- If $\pi = \delta_u$ degenerates to a point mass at u, then formally

$$\tilde{b}(y, \delta_u) = b(y, u),$$

so we recover the classical drift.

Exploratory Volatility

Definition 6.2

Exploratory volatility For $y \in \mathbb{R}$ and $\pi \in P(U)$, the exploratory volatility is defined by

$$\tilde{\sigma}(y, \pi) := \sqrt{\int_U \sigma^2(y, u)\pi(u)\, du}.$$

- Interpretation:
  - $\sigma(y, u)$ is the classical volatility under action u,
  - $\sigma^2(y, u)$ contributes to the local variance,
  - $\tilde{\sigma}(y, \pi)$ is chosen so that

$$\tilde{\sigma}^2(y, \pi) = \int_U \sigma^2(y, u)\pi(u)\, du.$$

- Thus the exploratory diffusion matches the **averaged local variance** produced by repeated randomized actions.
- This explains why the square root appears in the definition.

The Set $P(U)$ of Control Densities

- Here
$$P(U)$$
denotes the set of density functions of probability measures on U that are absolutely continuous with respect to Lebesgue measure.
- Concretely, $\pi \in P(U)$ means:
$$\pi(u) \geq 0 \quad \text{for all } u \in U, \qquad \int_U \pi(u)\, du = 1.$$
- So each $\pi \in P(U)$ is a probability density over the action space.
- Example 1: if $U = [0, 1]$, then
$$\pi(u) = 1, \qquad u \in [0, 1],$$
is the uniform density.
- Example 2: if $U = \mathbb{R}$, then a Gaussian density
$$\pi(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(u-\mu)^2}{2\sigma^2}\right)$$
belongs to $P(U)$.

Exploratory Reward

Definition 6.3

Exploratory reward For $y \in \mathbb{R}$ and $\pi \in P(U)$, define

$$\tilde{r}(y, \pi) := \int_U r(y, u)\pi(u)\, du.$$

- This is the natural analogue of the exploratory drift:
  - $r(y, u)$ is the classical reward,
  - $\tilde{r}(y, \pi)$ is the reward averaged over the randomized action.
- The repeated-learning argument gives the heuristic limit

$$\frac{1}{N}\sum_{i=1}^{N} e^{-\rho t} r(x_t^i, u_t^i)\Delta t \xrightarrow{\text{a.s.}} \mathbb{E}\left[e^{-\rho t}\int_U r(X_t^\pi, u)\pi_t(u)\, du\, \Delta t\right].$$

- Therefore, once exploration is modeled by a control distribution, the reward function must also be replaced by its averaged version.

Exploration Modifies Both Dynamics and Reward

- After introducing a control distribution π_t, the classical ingredients

$$b(y,u), \qquad \sigma(y,u), \qquad r(y,u)$$

  are replaced by

$$\tilde{b}(y,\pi), \qquad \tilde{\sigma}(y,\pi), \qquad \tilde{r}(y,\pi).$$

- So exploration changes the model in **two** ways:
  - it changes the **state dynamics**,
  - it changes the **reward functional**.
- This is why continuous-time RL with exploration cannot be handled by simply inserting random actions into the classical model.
- Instead, the model itself must be reformulated at the level of:
  - the drift,
  - the volatility,
  - the reward.

Connection with Relaxed Stochastic Control

- The exploratory formulation is consistent with the classical notion of **relaxed control** in stochastic control theory.
- In classical control:

$$u_t \in U.$$

- In relaxed/exploratory control:

$$\pi_t(\cdot) \in P(U).$$

- Thus the control variable is enlarged from a point in the action space to a probability distribution over the action space.
- Important conceptual point:
  - in the control literature, relaxed controls are often introduced to help with **existence** of optimal controls,
  - Here, the same mathematical structure is motivated by **learning and exploration in RL**.
- So the formulations are mathematically consistent, but their interpretations are different.

Why Entropy Regularization Is Needed

- We enlarged the control space from classical controls

$$u_t \in U$$

  to control distributions

$$\pi_t(\cdot) \in P(U).$$

- However, if the model were fully known, then there would be no need for learning or experimentation.
- In that case, the optimizer would select one best action only, so the optimal control distribution would collapse to a Dirac measure:

$$\pi_t = \delta_{u_t}.$$

- For any test function f,

$$\int_U f(u)\, \delta_{u_t}(du) = f(u_t).$$

- Therefore, without an additional incentive, the exploratory formulation would reduce to the classical stochastic control problem.

Why Entropy Regularization Is Needed

- A Dirac measure means:
  - all probability mass is concentrated on one action,
  - no randomization remains,
  - hence there is no genuine exploration.
- But in reinforcement learning, we want the controller to keep trying different actions while learning.
- Therefore we need a mechanism that rewards **non-degenerate** control distributions.
- This is exactly why we introduce **entropy regularization**.

Measuring Exploration by Shannon Differential Entropy

Definition 6.4

Shannon differential entropy For a density $\pi \in P(U)$, define

$$\mathcal{H}(\pi) := -\int_U \pi(u) \ln \pi(u)\, du.$$

- The entropy $\mathcal{H}(\pi)$ measures how **spread out** the control distribution is.
- Interpretation:
  - small entropy $\Longrightarrow$ the distribution is highly concentrated,
  - large entropy $\Longrightarrow$ the distribution is more spread out.
- Therefore:

$$\text{larger entropy} \Longleftrightarrow \text{more exploration.}$$

- Extreme cases:
  - **Dirac measure:** no exploration,
  - **more diffuse distribution:** greater exploration.

Entropy: Simple Examples

- **Example 1: Uniform density on** $[0,1]$

$$\pi(u) = 1, \qquad u \in [0,1].$$

Then

$$\mathcal{H}(\pi) = -\int_0^1 1 \cdot \ln 1 \, du = 0.$$

- **Example 2: Gaussian density on** $\mathbb{R}$

$$\pi(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(u-\mu)^2}{2\sigma^2}\right).$$

Its entropy is

$$\mathcal{H}(\pi) = \frac{1}{2}\ln(2\pi e\sigma^2).$$

- Hence, in the Gaussian case,

$$\sigma^2 \uparrow \quad \Longrightarrow \quad \mathcal{H}(\pi) \uparrow .$$

Entropy-Regularized Objective Function

- To balance exploitation and exploration, we introduce the entropy-regularized objective

$$V(x) := \sup_{\pi \in \mathcal{A}(x)} \mathbb{E}\Big[\int_0^\infty e^{-\rho t} \Big(\int_U r(X_t^\pi, u) \pi_t(u)\, du - \lambda \int_U \pi_t(u) \ln \pi_t(u)\, du \Big)\, dt \;\Big|\; X_0^\pi = x \Big].$$

- The first term,

$$\int_U r(X_t^\pi, u) \pi_t(u)\, du,$$

  is the **exploratory reward**.
- The second term,

$$-\lambda \int_U \pi_t(u) \ln \pi_t(u)\, du = \lambda\, \mathcal{H}(\pi_t),$$

  is the **entropy bonus**.

Entropy-Regularized Objective Function

- The two terms above can now be interpreted as:
  - the **exploratory reward**

    $$\int_U r(X_t^\pi, u)\pi_t(u)\, du$$

    acts as the **exploitation term**,
  - the **entropy bonus**

    $$\lambda\, \mathcal{H}(\pi_t)$$

    acts as the **exploration term**.
- Therefore, maximizing $V(x)$ means balancing exploitation and exploration.

Interpretation of the Exploration Weight λ

- The constant
$$\lambda > 0$$
is an exogenous **exploration weight parameter**.
- It determines the trade-off between:
  - **exploitation:** maximizing the original reward,
  - **exploration:** encouraging randomness through entropy.
- If λ is small, then the entropy term has little influence:
$$\text{policy is closer to pure exploitation.}$$
- If λ is large, then entropy matters more:
$$\text{policy becomes more exploratory.}$$
- As
$$\lambda \to 0,$$
the exploration incentive disappears, and we expect the model to approach the classical stochastic control problem.

Admissible Exploratory Control Distributions

Definition 6.5

An admissible exploratory control is a process

$$\pi = \{\pi_t,\ t \geq 0\}$$

satisfying the following properties:

(i) for each $t \geq 0$,

$$\pi_t \in P(U) \quad \text{a.s.},$$

(ii) for each $A \in \mathcal{B}(U)$, the process

$$\left\{ \int_A \pi_t(u)\, du,\ t \geq 0 \right\}$$

is $\{\mathcal{F}_t\}$-progressively measurable,

Admissible Exploratory Control Distributions

Definition 6.5

(iii) the exploratory SDE

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\, dt + \tilde{\sigma}(X_t^\pi, \pi_t)\, dW_t, \qquad X_0^\pi = x,$$

has a unique strong solution $X^\pi = \{X_t^\pi,\ t \geq 0\}$,

(iv) the associated reward functional is well-defined, i.e.

$$\mathbb{E}\left[\int_0^\infty e^{-\rho t} \left|\tilde{r}(X_t^\pi, \pi_t) - \lambda \int_U \pi_t(u) \ln \pi_t(u)\, du\right| dt\right] < \infty.$$

We denote the set of such admissible controls by

$$\mathcal{A}(x).$$

Understanding Condition (i): $\pi_t \in P(U)$

- The condition
$$\pi_t \in P(U) \quad \text{a.s.}$$
means that at each time t, π_t is a probability density on the action space U.
- So for each t,
$$\pi_t(u) \geq 0, \qquad \int_U \pi_t(u)\, du = 1.$$
- Example: if $U = \mathbb{R}$, then a Gaussian density
$$\pi_t(u) = \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{(u-\mu_t)^2}{2\sigma_t^2}\right), \qquad \sigma_t^2 > 0,$$
belongs to $P(U)$.
- Interpretation:
  - in the classical setting, the controller chooses one action u_t,
  - here the controller chooses a full density π_t.

Understanding Condition (ii): Progressive Measurability

- For each Borel set $A \subseteq U$, the quantity

$$\int_A \pi_t(u)\, du$$

  is the probability assigned by π_t to the action set A.
- The requirement

$$\left\{ \int_A \pi_t(u)\, du,\ t \geq 0 \right\} \text{ is progressively measurable}$$

  means that these probabilities are determined using only information available up to time t.
- So the controller cannot use future information when choosing the action distribution.
- This is the distribution-valued analogue of adaptedness for ordinary controls.

Understanding Condition (iii): Unique Strong Solution

- Recall the exploratory SDE:

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\, dt + \tilde{\sigma}(X_t^\pi, \pi_t)\, dW_t, \qquad X_0^\pi = x.$$

- A **strong solution** means:
  - the probability space is fixed,
  - the Brownian motion W_t is fixed,
  - the solution X_t^π is adapted to the same filtration.
- This is natural in control and RL because:
  - the policy must use current information,
  - the state process should evolve under the same observed randomness.
- So condition (iii) guarantees that the state process under π is mathematically well-defined.

Understanding Condition (iv): Finite Reward Functional

- Even if π_t is a valid density and the SDE has a solution, we still need the objective functional to be finite.
- That is why we require

$$\mathbb{E}\left[\int_0^\infty e^{-\rho t}\left|\tilde{r}(X_t^\pi, \pi_t) - \lambda \int_U \pi_t(u) \ln \pi_t(u)\, du\right| dt\right] < \infty.$$

- This ensures that the value

$$V(x) = \sup_{\pi \in \mathcal{A}(x)} \mathbb{E}[\cdots]$$

  is well-defined.
- In words:
  - condition (i): π_t is a probability density,
  - condition (ii): π_t is information-adapted,
  - condition (iii): the state dynamics make sense,
  - condition (iv): the reward functional makes sense.

Feedback Control Distributions

- We also consider **feedback control distributions**.
- This is the distribution-valued analogue of classical feedback control.
- Classical feedback control:

$$u_t = \mathbf{u}(X_t).$$

- Feedback control distribution:

$$\pi_t = \Pi(X_t^\pi),$$

where

$$\Pi : \mathbb{R} \to P(U).$$

- So:
  - Π is the deterministic rule assigning a density to each state,
  - π_t is the actual density used at time t.

Feedback Exploratory Dynamics

- When a feedback control distribution is used, the exploratory state process satisfies

$$dX_t^\pi = \tilde{b}\big(X_t^\pi, \Pi(X_t^\pi)\big)\,dt + \tilde{\sigma}\big(X_t^\pi, \Pi(X_t^\pi)\big)\,dW_t, \qquad X_0^\pi = x. \quad (1)$$

- This is the feedback version of the general exploratory SDE.
- Compare:
  - general open-loop exploratory control:

$$dX_t^\pi = \tilde{b}(X_t^\pi, \pi_t)\,dt + \tilde{\sigma}(X_t^\pi, \pi_t)\,dW_t,$$

  - feedback exploratory control:

$$\pi_t = \Pi(X_t^\pi),$$

    so the coefficients become state-dependent through the feedback map Π.
- This is exactly parallel to classical feedback control, but now the rule outputs a density rather than a single action.

Recap: Main Message

- Exploration cannot be represented merely by enlarging the control set to $P(U)$.
- Without an incentive to explore, the optimizer would collapse to Dirac measures:

$$\pi_t = \delta_{u_t}.$$

- Therefore one can add Shannon entropy as a regularization term:

$$\mathcal{H}(\pi) = -\int_U \pi(u) \ln \pi(u)\, du.$$

- The entropy-regularized objective balances:
  - **reward** $\Longrightarrow$ exploitation,
  - **entropy** $\Longrightarrow$ exploration.
- The admissible control class $\mathcal{A}(x)$ ensures that:
  - the control distributions are well-defined,
  - the exploratory state process exists,
  - the optimization problem is mathematically meaningful.

What Changes Now?

- We replaced the classical control

$$u_t \in U$$

by a control distribution

$$\pi_t(\cdot) \in P(U).$$

- We also replaced the classical coefficients by their exploratory counterparts:

$$\tilde{b}(x,\pi) = \int_U b(x,u)\pi(u)\,du,$$

$$\tilde{\sigma}(x,\pi) = \left(\int_U \sigma^2(x,u)\pi(u)\,du\right)^{1/2},$$

$$\tilde{r}(x,\pi) = \int_U r(x,u)\pi(u)\,du.$$

What Changes Now?

- Finally, we added Shannon entropy regularization:

$$\mathcal{H}(\pi) = -\int_U \pi(u) \ln \pi(u)\, du.$$

- So the next question is natural:

  What is the Bellman/HJB equation for this exploratory control problem?

Classical HJB Reminder

- For a classical infinite-horizon discounted control problem, the value function v satisfies

$$\rho v(x) = \max_{u \in U} \left\{ r(x,u) + b(x,u)v'(x) + \frac{1}{2}\sigma^2(x,u)v''(x) \right\}.$$

- Interpretation of the terms:
  - $r(x,u)$: immediate reward,
  - $b(x,u)v'(x)$: effect of the drift on future value,
  - $\frac{1}{2}\sigma^2(x,u)v''(x)$: effect of the diffusion on future value.
- This is the continuous-time analogue of the Bellman optimality equation from discrete-time RL.
- In the classical setting, the control variable is a **point**:

$$u \in U.$$

Why Classical HJB Is Not Enough Here

- In our exploratory formulation, the controller does not directly choose one action u.
- Instead, the controller chooses a **density**

$$\pi \in P(U),$$

  and the action is sampled from that density.
- Therefore the classical HJB must be modified in two ways:
  - maximize over π, not over u,
  - include the entropy reward term

$$\lambda \mathcal{H}(\pi).$$

- So the exploratory HJB should answer:

  At state x, which control distribution π gives the best balance between reward, future value, and exploration?

The Exploratory HJB Equation

- Applying Bellman's principle to the exploratory problem gives

$$\rho v(x) = \max_{\pi \in P(U)} \Big(\tilde{r}(x,\pi) - \lambda \int_U \pi(u) \ln \pi(u)\, du + \tilde{b}(x,\pi) v'(x) + \frac{1}{2} \tilde{\sigma}^2(x,\pi) v''(x) \Big).$$

- This is the direct analogue of the classical HJB, but now the control variable is a density.
- In words, the right-hand side contains:
  - exploratory reward,
  - entropy bonus,
  - drift contribution to continuation value,
  - diffusion contribution to continuation value.
- This is the key PDE for continuous-time RL under exploration.

What Are We Going To Do Next?

- The exploratory HJB is now written, but it still contains a difficult maximization over all densities $\pi \in P(U)$.
- So the next goal is:

  solve the constrained optimization problem inside the HJB.

- Concretely, we will:
  1. rewrite the HJB in integral form;
  2. isolate the quantity that depends on the control variable u;
  3. optimize over π subject to

$$\pi(u) \geq 0, \qquad \int_U \pi(u)\, du = 1;$$

  4. obtain the optimal feedback control distribution $\pi^*(u; x)$;
  5. substitute π^* back into the HJB.
- This is the continuous-time analogue of deriving a softmax policy in discrete-time RL.

Integral Form of the Exploratory HJB

- Using the definitions of $\tilde{r}, \tilde{b}, \tilde{\sigma}$, we can rewrite the HJB as

$$\rho v(x) = \max_{\pi \in P(U)} \int_U \Big(r(x,u) - \lambda \ln \pi(u) + b(x,u) v'(x) + \frac{1}{2}\sigma^2(x,u) v''(x) \Big) \pi(u)\, du.$$

- This form is better for computation because the optimization variable π appears explicitly.
- Define the score

$$G(x,u) := r(x,u) + b(x,u) v'(x) + \frac{1}{2}\sigma^2(x,u) v''(x).$$

- Then the HJB becomes

$$\rho v(x) = \max_{\pi \in P(U)} \int_U \big(G(x,u) - \lambda \ln \pi(u) \big) \pi(u)\, du.$$

Optimization Over Densities

- We now solve

$$\max_{\pi \in P(U)} \int_U \big(G(x,u) - \lambda \ln \pi(u)\big)\pi(u)\, du$$

subject to

$$\pi(u) \geq 0, \qquad \int_U \pi(u)\, du = 1.$$

- Using a Lagrange multiplier for the normalization constraint, the first-order condition gives

$$\ln \pi(u) = \frac{1}{\lambda} G(x,u) - \text{constant}.$$

- Therefore the optimizer must satisfy

$$\pi(u) \propto \exp\left(\frac{1}{\lambda} G(x,u)\right).$$

- After normalization, this yields the optimal feedback control distribution.

Optimal Feedback Control Distribution

Definition 6.6

Optimal exploratory feedback control The optimal control distribution is

$$\pi^*(u;x) = \frac{\exp\left(\frac{1}{\lambda}\left[r(x,u) + b(x,u)v'(x) + \frac{1}{2}\sigma^2(x,u)v''(x)\right]\right)}{\int_U \exp\left(\frac{1}{\lambda}\left[r(x,\tilde{u}) + b(x,\tilde{u})v'(x) + \frac{1}{2}\sigma^2(x,\tilde{u})v''(x)\right]\right)\, d\tilde{u}}.$$

- This is a Gibbs / Boltzmann / softmax-type distribution.
- Actions with larger score receive larger probability density.
- The parameter λ controls how concentrated or spread out the distribution is.

Interpretation of $\pi^*(u; x)$

- The quantity

$$r(x,u) + b(x,u)v'(x) + \frac{1}{2}\sigma^2(x,u)v''(x)$$

  is the total score of action u at state x.
- It combines:
  - immediate reward,
  - drift effect on future value,
  - diffusion effect on future value.
- So $\pi^*(u; x)$ assigns more probability to actions that score better.
- But because of the entropy term, the controller does not collapse immediately to a single action.
- Thus:
  - exploitation favors high-score actions,
  - exploration keeps the policy spread out.

Reduced Exploratory HJB

- After substituting $\pi^*(u;x)$ back into the HJB, the control distribution disappears from the equation.
- The reduced HJB becomes

$$\rho v(x) = \lambda \ln\left(\int_U \exp\left(\frac{1}{\lambda}\Big[r(x,u) + b(x,u)v'(x) + \frac{1}{2}\sigma^2(x,u)v''(x)\Big]\right) du\right).$$

- This is a **soft HJB equation**.
- Compare:
  - classical HJB: hard maximum over u,
  - exploratory HJB: soft log-integral-exp over u.
- This is the form we will now analyze in the linear–quadratic case.

Bridge to the LQ Case

- Up to this point, the framework is completely general.
- The next question is:

  Can we solve this exploratory HJB explicitly?

- In general, the answer is difficult.
- But in the linear–quadratic setting:
  - the state dynamics are linear,
  - the reward is quadratic,
  - the HJB becomes tractable,
  - the optimal exploratory control turns out to be Gaussian.
- So the LQ case will give the first fully explicit answer to the continuous-time exploration problem.